

Nathaniel Murcin

603-303-2612 | nmurcin3@gatech.edu | linkedin.com/in/nathaniel-murcin/ | nmurcin.github.io

Education

Georgia Institute of Technology

B.S. Aerospace Engineering | B.S./M.S. Aerospace Engineering

Minor in Astrophysics

- Relevant Coursework: Propulsion, Thermodynamics, Structural Analysis, Aerodynamics, Aerothermodynamics

GPA: 4.00/4.00

B.S. Expected May 2027

M.S. Expected May 2028

Experience

Blue Origin

Fluids & Manufacturing Engineering Intern

Cape Canaveral, FL

May 2026 – Aug. 2026

Fluids Engineering

- Designed a cryogenic tank-farm P&ID for a critical-path ground-test facility, reducing component count ~60% versus heritage designs by downsizing, eliminating branches, and challenging requirements.
- Sized, selected, and validated cryogenic piping, valves, and relief devices using Cv, Darcy-Weisbach, line-sizing, and overpressure-relief analyses, then developed the BOM and coordinated supplier procurement.
- Performed preliminary hazard analysis for a major cryogenic fluid system, evaluating component failure modes and consequences against safeguards while documenting CONOPS for formal HAZOP review.

Lunar Manufacturing

- Designed 6 iterations of an adaptable gripping fixture in Creo, preventing part damage, slippage, and metal-chip FOD, while improving technician usability while reducing fastener count from 8 to 1; design passed PDR.
- Validated the fixture with hand calculations and Creo static-structural FEA, evaluating torque resistance, bolt shear/tension, preload, slip, tear-out, bending, and CTE effects against company yield and ultimate margin criteria.

Pursuit Aerospace

Engineering Intern

Newburyport, MA

Aug. 2025 – Dec. 2025

- Prototyped, designed, built, and deployed 15 quick-change dot-peen marking fixtures using GD&T and DFM, reducing average part load/unload time by 83%.
- Designed Go/No-Go gauges that reduced inspection time by 95%; broader one-piece-flow changes reduced time between completed parts by 55%.
- Developed an in-house precision cleaning process for a Rolls-Royce component, qualified for production use by Rolls-Royce and Pursuit Quality, cut lead time by 3 days and save \$1,000/part in outsourced cleaning and shipping.

Georgia Tech Experimental Rocketry (GTXR)

Fin Design & Manufacturing Lead

Atlanta, GA

Aug. 2023 – Present

- Led a structured fin design trade study evaluating materials, geometries, and manufacturing constraints, down-selecting feasible concepts for analytical feasibility, machinability, and engineering first principles.
- Co-developed and evaluated a thermoforming process for double-wedge PMI foam fin cores, using CNC-machined aluminum tooling and 3D metrology to achieve 0.28 mm average global geometric error.
- Designed, CAM programmed, and machined a modular four-tower fin-alignment jig; independent metrology verified all four fins within $<0.1^\circ$ of the 1.6° target.
- Designed and fabricated a flight-critical NPT electrical passthrough in 14 hours during launch integration after flight unit failure; hardware passed hydrostatic testing at $2\times$ MEOP, enabled successful flight.
- Designed and executed plate-twist testing apparatus to determine in-plane shear modulus of composite sandwich panels, verifying the ANSYS model within a 5% acceptance criterion for future fin development.

Ben T. Zinn Combustion Laboratory

Undergraduate Researcher

Atlanta, GA

Jan. 2026 – Present

- Supported reacting jets-in-crossflow experiments by assembling regulated multi-gas delivery hardware and executing established MATLAB/PIV post-processing workflows.

Publications

- **Co-author & presenter:** *Shear Modulus Testing in Composite Sandwich Panels*, AIAA Regional Student Conference Proceedings, 2024 (AIAA 2024-85690).
- **Co-author & presenter:** *Viability of Thermoformed Closed-Cell Foam Cores for Carbon Fiber Sandwich Rocket Fins*, AIAA Regional Student Conference Proceedings, 2026 (AIAA 2026-112793).

Technical Skills

Analysis: MATLAB, P&ID design, cryogenic sizing & safety, Structural FEA, bolted joint, flutter analysis,

Hardware & Manufacturing: GD&T, DFM, 3D printing, composite layups, Instron, CNC mill, lathe, waterjet